

F.T. $\frac{2 \sin(\omega T_1)}{\omega} = 2 T_1 \text{Sinc}\left(\frac{\omega T_1}{\pi}\right)$

$f(t - t_0) \leftrightarrow e^{-j\omega t_0} F(j\omega)$

$t_0 = \frac{3}{2}$ $e^{-j\omega \frac{3}{2}} \cdot \frac{2 \sin(\frac{\omega}{2})}{\omega}$

$t_0 = -\frac{3}{2}$ $e^{+j\omega \frac{3}{2}} \cdot \frac{2 \sin(\frac{\omega}{2})}{\omega}$

$F(j\omega) = 2 \cdot \frac{2 \sin(\frac{\omega}{2})}{\omega} \left(\frac{e^{+j\omega \frac{3}{2}} + e^{-j\omega \frac{3}{2}}}{2} \right)$

$= 4 \frac{\sin(\frac{\omega}{2})}{\omega} \cos\left(\frac{3\omega}{2}\right)$

Partial-Fraction Expansion (PFE)

(I) $G(s) = \frac{A}{(s+1)} + \frac{B}{(s+3)}$ s could be complex variable
 $s = j\omega$

$G(s) = \frac{s+2}{s^2 + 4s + 3} = \frac{s+2}{(s+1)(s+3)}$ (I)

To find A & B by PFE

+ To find A:

• multiply both sides of (I) by $(s+1)$

$(s+1) \frac{s+2}{(s+1)(s+3)} = (s+1) \frac{A}{(s+1)} + (s+1) \frac{B}{(s+3)}$

• Evaluate at $s = -1$

$A = \frac{-1+2}{-1+3} = \frac{1}{2}$

When $s = -1$ this becomes 0

+ To find B:

• multiply by $(s+3)$ and evaluate it for $s = -3$

$B = \frac{-3+2}{-3+1} = \frac{1}{2}$

$$\therefore G(s) = \frac{s+2}{(s+1)(s+3)} = \frac{1/2}{s+1} + \frac{1/2}{s+3}$$

in general:

$$H(s) = \frac{b_{n-1}s^{n-1} + b_{n-2}s^{n-2} + \dots + b_1s + b_0}{s^n + a_{n-1}s^{n-1} + \dots + a_1s + a_0}$$

n = درجه چندجبرایی مخرج

$n-1$ = درجه چندجبرایی صورت

وقتی ریشه‌های چندجبرایی مخرج متمایز باشند، شبیه به مثال بالا خواهد بود

$$H(s) = \frac{A_1}{s-p_1} + \frac{A_2}{s-p_2} + \dots + \frac{A_n}{s-p_n}$$

اگر مخرج ریشه‌های تکراری داشته باشد: مثلاً

$$H(s) = \frac{b_2s^2 + b_1s + b_0}{(s-p_1)^2(s-p_2)} \rightarrow \text{چندجبرایی صورت } n=2$$

$\underbrace{\hspace{10em}}$
ریشه تکراری

در این صورت PFE به این صورت می‌باشد:

$$H(s) = \frac{A_{11}}{(s-p_1)} + \frac{A_{12}}{(s-p_1)^2} + \frac{A_{21}}{(s-p_2)}$$

To find A_{12} :

$$A_{12} = (s-p_1)^2 H(s) \Big|_{s=p_1}$$

To find A_{21} :

$$A_{21} = (s-p_2) H(s) \Big|_{s=p_2}$$

To find A_{11} :

$$A_{11} = \frac{d}{ds} \left[(s-p_1)^2 H(s) \right] \Big|_{s=p_1}$$

Example

$$Y(s) = \frac{s+2}{(s+1)^2(s+3)} = \frac{A_{11}}{s+1} + \frac{A_{12}}{(s+1)^2} + \frac{A_{21}}{s+3}$$

$$A_{11} = \frac{d}{ds} \left((s+1)^2 Y(s) \right) \Big|_{s=-1} = \frac{1}{4}$$

$$A_{12} = (s+1)^2 Y(s) \Big|_{s=-1} = \frac{1}{2}$$

$$A_{21} = (s+3)Y(s) \Big|_{s=-3} = -\frac{1}{4}$$

$$Y(s) = \frac{s+2}{(s+1)^2(s+3)} = \frac{\frac{1}{4}}{s+1} + \frac{\frac{1}{2}}{(s+1)^2} + \frac{-\frac{1}{4}}{(s+3)}$$

Example: CTFT

Find the F.T. of $x(t)$:

$$x(t) = e^{-a|t|} (1 + a|t|)$$

Solution:

$$\begin{aligned} x(t) &= e^{-a|t|} + a|t| e^{-a|t|} \\ &= e^{-a|t|} + a \begin{cases} t e^{-at} & t > 0 \\ -t e^{at} & t < 0 \end{cases} \\ &= e^{-a|t|} + a \underbrace{t e^{-at} u(t)}_{f(t)} - a t e^{at} u(-t) \\ &= e^{-a|t|} + a f(t) + a f(-t) \end{aligned} \quad f(t) = t e^{-at} u(t)$$

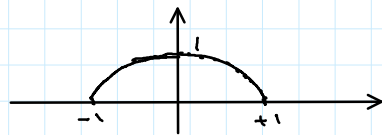
take the F.T. of $x(t)$:

$$X(j\omega) = \frac{2a}{a^2 + \omega^2} + a F(j\omega) + a F(-j\omega)$$

$$f(t) = t e^{-at} u(t) \xleftrightarrow{\text{F.T.}} F(j\omega) = \frac{1}{(a + j\omega)^2}$$

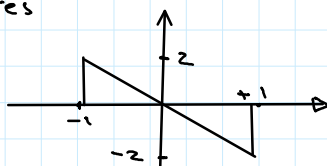
$$\begin{aligned} X(j\omega) &= \frac{2a}{a^2 + \omega^2} + \frac{a}{(a + j\omega)^2} + \frac{a}{(a - j\omega)^2} \\ &= \frac{4a^3}{(a^2 + \omega^2)^2} \end{aligned}$$

Ex Find F.T. of $x(t) = 1 - t^2 \quad |t| < 1$



Solution: use properties

$$y_1(t) = \frac{dx(t)}{dt}$$



$$\longleftrightarrow Y_1(j\omega) = (j\omega) X(j\omega)$$

$$y_2(t) = \frac{d^2 x(t)}{dt^2}$$

$$Y_2(j\omega) = (j\omega)^2 X(j\omega)$$

$$= 2\delta(t-1) + 2\delta(t+1) - 2\pi\left(\frac{t}{2}\right) \quad (I)$$

$$\pi(t) \xleftrightarrow{F.T.} \frac{2\sin(\frac{\omega}{2})}{\omega}$$

$$f(at) = \frac{1}{|a|} F\left(\frac{\omega}{a}\right)$$

$$\pi\left(\frac{t}{2}\right) \xleftrightarrow{F.T.} \frac{2\sin(\omega)}{\omega}$$

From Eq (I)

$$Y_2(j\omega) = \underbrace{2e^{-j\omega} + 2e^{+j\omega}}_{4\cos(\omega)} - 2 \frac{2\sin(\omega)}{\omega}$$

$$= 4\cos(\omega) - 4 \frac{\sin(\omega)}{\omega}$$

$$X(j\omega) = \frac{Y_2(j\omega)}{-\omega^2} = \frac{4\cos(\omega) - 4 \frac{\sin(\omega)}{\omega}}{-\omega^2}$$

$$= \frac{4\sin(\omega)}{\omega^3} - \frac{4\cos(\omega)}{\omega^2}$$

Ex

Evaluate:

$$\int_{-\infty}^{+\infty} X(\omega) \cdot \underbrace{\frac{2\sin(\omega)}{\omega} e^{j2\omega}}_{Y^*(j\omega)} d\omega \quad (I)$$



Solution:

Use Parseval theorem:

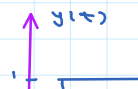
$$\int_{-\infty}^{+\infty} |X(j\omega)|^2 d\omega = 2\pi \int_{-\infty}^{+\infty} |x(t)|^2 dt$$

in general:

$$\int_{-\infty}^{+\infty} X(j\omega) \cdot Y^*(j\omega) d\omega = 2\pi \int_{-\infty}^{+\infty} x(t) y^*(t) dt$$

in eq. (I)

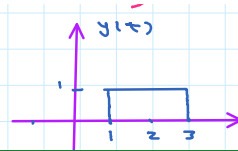
$$Y^*(j\omega) = \frac{2\sin(\omega)}{\omega} e^{j2\omega}$$



0

$$Y^*(j\omega) = \frac{2 \sin(\omega)}{\omega} e^{j2\omega}$$

$$Y(j\omega) = \frac{2 \sin(\omega)}{\omega} e^{-j2\omega}$$



Recall :

* $\xleftrightarrow{F.T.} \frac{2 \sin(\omega T_1)}{\omega} = 2 T_1 \text{sinc}\left(\frac{\omega T_1}{\pi}\right)$

* $\frac{\sin(\omega t)}{\pi t} = \frac{W}{\pi} \text{sinc}\left(\frac{\omega t}{\pi}\right) \longleftrightarrow$ ω

Exc Find IFT of the following : $H(j\omega) = \frac{(\sin^2(3\omega)) \cos(\omega)}{\omega^2}$

Sol. $h(t) = f^{-1}\{H(j\omega)\} = f^{-1}\left\{\left(\frac{\sin(3\omega)}{\omega}\right)^2 \cos(\omega)\right\}$

Recall :

$\longleftrightarrow 2 \frac{\sin(\omega T_1)}{\omega} = 2 T_1 \frac{\sin(\pi \omega T_1 / \pi)}{\pi \omega T_1 / \pi}$

$= 2 T_1 \text{sinc}\left(\frac{\omega T_1}{\pi}\right)$

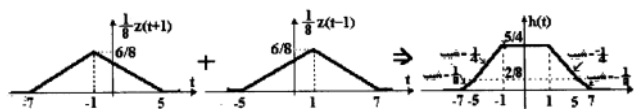
$T_1 = 3$

$\therefore \frac{1}{4} \left(\pi\left(\frac{t}{6}\right) * \pi\left(\frac{t}{6}\right) \right) \xleftrightarrow{F.T.} \left(\frac{\sin(3\omega)}{\omega} \right)^2$

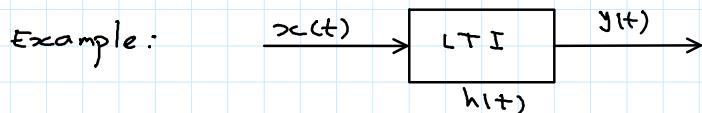
So : $f^{-1}\left\{\left(\frac{\sin(3\omega)}{\omega}\right)^2 \cos(\omega)\right\} = \frac{1}{4} \left(\pi\left(\frac{t}{6}\right) * \pi\left(\frac{t}{6}\right) \right) * \frac{1}{2} \left\{ \delta(t+1) + \delta(t-1) \right\}$

$\xleftrightarrow{F.T.} 4 \left(\frac{\sin(3\omega)}{\omega} \right)^2$

$\therefore \frac{1}{8} z(t) * \left\{ \delta(t+1) + \delta(t-1) \right\} = \frac{1}{8} z(t+1) + \frac{1}{8} z(t-1)$



$$h(t) = \begin{cases} \frac{5}{4} & ; |t| < 1 \\ -\frac{|t|}{4} + \frac{3}{2} & ; 1 \leq |t| \leq 5 \\ -\frac{|t|}{8} + \frac{7}{8} & ; 5 \leq |t| < 7 \\ 0 & ; \text{otherwise} \end{cases}$$

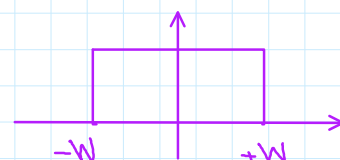


if & $x(t) = \cos(2\pi t) + \sin(6\pi t)$
 $h(t) = \frac{\sin(4\pi t)}{\pi t}$ Find $y(t)$.

Solution: Solve it in Fourier domain $Y(j\omega) = X(j\omega) \cdot H(j\omega)$

use:

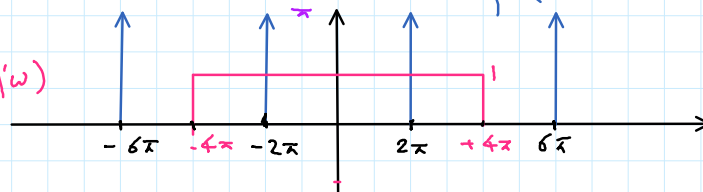
$$\frac{\sin(Wt)}{\pi t} \longleftrightarrow$$



$$\cos(\omega_0 t) \longleftrightarrow \pi (\delta(\omega - \omega_0) + \delta(\omega + \omega_0))$$

$$\sin(\omega_0 t) \longleftrightarrow \frac{\pi}{j} (\delta(\omega - \omega_0) - \delta(\omega + \omega_0))$$

$\therefore |X(j\omega)|$ & $H(j\omega)$

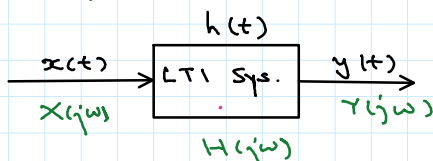


$\therefore y(t) = \cos(2\pi t)$

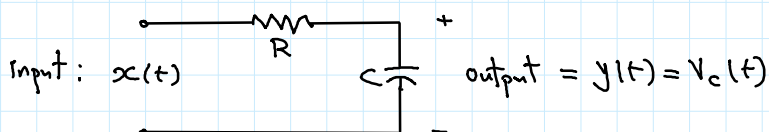
Part (B): Solve this prob. with

$$h(t) = \frac{\sin(4\pi t) \cdot \sin(8\pi t)}{\pi t^2}$$

FT Properties: Convolution:



$$Y(j\omega) = X(j\omega) \cdot H(j\omega)$$



Recall: $\frac{dy(t)}{dt} + \frac{1}{RC} y(t) = \frac{1}{RC} x(t)$

use F.T. $j\omega Y(j\omega) + \frac{1}{RC} Y(j\omega) = \frac{1}{RC} X(j\omega)$

$$\left(\frac{1}{RC} + j\omega\right) Y(j\omega) = \frac{1}{RC} X(j\omega)$$

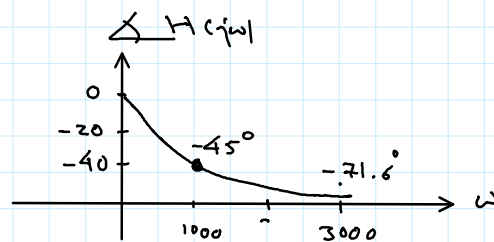
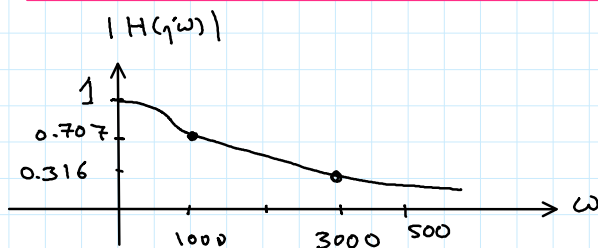
$$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)} = \frac{\frac{1}{RC}}{j\omega + \frac{1}{RC}}$$

Freq. Response
تأثير التردد

مثال ϵ^- $|H(j\omega)| = \frac{\frac{1}{RC}}{\sqrt{\omega^2 + \left(\frac{1}{RC}\right)^2}}$

$$\angle = -\tan^{-1}(\omega RC)$$

For above example: Let $\frac{1}{RC} = 1000$



if

$x(t) = A \cos(\omega_0 t + \theta)$ is used as input

then, output would be,

$$y(t) = A \cdot |H(j\omega_0)| \cos(\omega_0 t + \theta + \angle H(j\omega_0))$$

so,

if $\omega_0 = 0$ $x(t) = A$ constant source

then

$$y(t) = A \underbrace{(1)}_{|H(0)|} \cos(0t + \theta + 0) = A \cos(\theta)$$

if $\omega_0 = 1000$ rad/s

$$y(t) = A \underbrace{(0.707)}_{|H(1000)|} \cos(\underbrace{1000t}_{\text{Same freq.}} + \theta + \underbrace{45^\circ}_{\text{phase shift}})$$

if $\omega_0 = 3000$ rad/s

$$y(t) = A (0.316) \cos(3000t + \theta - 71.6^\circ)$$

∴ the RC circuit ,

for low frequency $\Rightarrow$ less amplitude ,
but same Freq.

as

$\omega_0 \rightarrow \infty$ the amplitude of output is decreased and
approach to 0

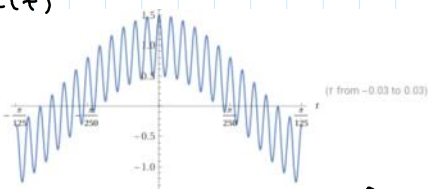
it is a Low-pass Filter.

Now: Let

$$x(t) = A_1 \cos(\omega_1 t + \theta_1) + A_2 \cos(\omega_2 t + \theta_2)$$

$$\omega_2 \gg \omega_1$$

$x(t)$

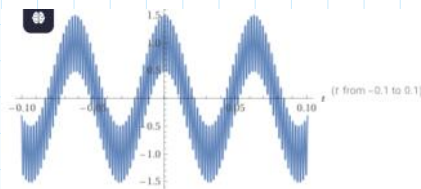


$$\omega_1 = 100$$

$$\omega_2 = 3000$$

$$A_1 = 1$$

$$A_2 = 0.5$$

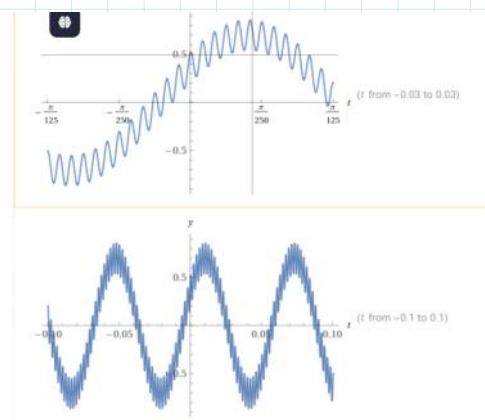


then the output would be,

$$y(t) = A_1 |H(\omega_1)| \cos(\omega_1 t + \theta_1 + \angle H(\omega_1))$$

$$+ A_2 |H(\omega_2)| \cos(\omega_2 t + \theta_2 + \angle H(\omega_2))$$

$$= 0.707 \cos(1000t + \theta_1 - 45^\circ) + (0.5)(0.316) \cos(3000t + \theta_2 - 71.6^\circ)$$



Last Note :

$$e^{j\omega_0 t} \xleftrightarrow{\text{F.T.}} 2\pi \delta(\omega - \omega_0)$$

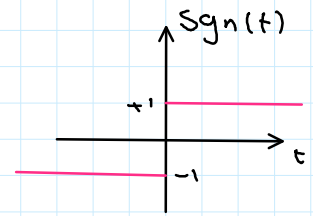
if $\omega_0 = 0$
then

$$1 \xleftrightarrow{\text{F.T.}} 2\pi \delta(\omega)$$

or

$$K \xleftrightarrow{\text{F.T.}} 2\pi K \delta(\omega)$$

Let: signum function: $\text{sgn}(t) = \begin{cases} -1 & t < 0 \\ +1 & t > 0 \end{cases}$

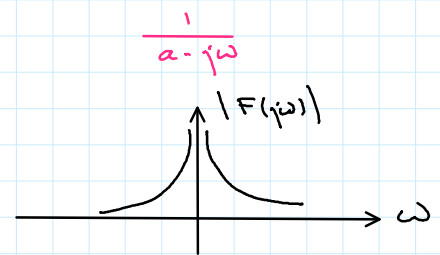


Also: $\text{sgn}(t) = u(t) - u(-t)$

$$\text{sgn}(t) = \lim_{a \rightarrow 0} \left(e^{-at} u(t) - e^{at} u(-t) \right)$$

$$\text{F.T.} \{ \text{sgn}(t) \} = \lim_{a \rightarrow 0} \left(\int_0^{\infty} e^{-at} e^{-j\omega t} dt - \int_{-\infty}^0 e^{at} e^{-j\omega t} dt \right)$$

$$= \lim_{a \rightarrow 0} \frac{-2j\omega}{\omega^2 + a^2} = \frac{2}{j\omega}$$



So:

$$\text{sgn}(t) \xleftrightarrow{\text{F.T.}} \frac{2}{j\omega}$$

Also:

$$u(t) = \frac{1}{2} + \frac{1}{2} \text{sgn}(t)$$

based on previous results:

$$\text{F.T.} \{ u(t) \} = \text{F.T.} \left\{ \frac{1}{2} + \frac{1}{2} \text{sgn}(t) \right\}$$

$$= \left(\frac{1}{2} \right) 2\pi \delta(\omega) + \frac{1}{2} \left(\frac{2}{j\omega} \right)$$

$$= \pi \delta(\omega) + \frac{1}{j\omega}$$

integration Property :

$$x(t) * u(t) = \int_{-\infty}^t x(\tau) d\tau$$

its F.T. is :

$$= X(j\omega) \cdot \text{F.T}\{u(t)\}$$

$$= X(j\omega) \cdot \left\{ \pi \delta(\omega) + \frac{1}{j\omega} \right\}$$

$$= \pi X(0) \delta(\omega) + \frac{X(j\omega)}{j\omega}$$